

Data-driven accident consequence assessment on urban gas pipeline network based on machine learning

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ABSTRACT

The accidents in Urban Pipeline Network (UPN) may cause enormous economic loss and serious threats to society and environment. The daily operation and maintenance of UPN is usually associated with many aspects of data. How to take full advantage of these multi-source data in combination with advanced data mining techniques to assess the post-event risk of pipeline accidents is of great significance to management of resilient urban systems. This work first summarizes the factors affecting accident consequence of gas UPN and establishes the risk evaluation indicators. A traditional risk assessment model based on the Kent index method and the analytic hierarchy process is then employed to determine the relative risk value of each pipeline. To reduce the dependency on experts' subjective judgements or calculation of probability events in a Bayes decision procedure, a data-driven model based on graph embedding and clustering algorithm is proposed. The Graph Convolutional Network (GCN) technique is used to extract the topological features of pipeline network as a complement to the common attribute features considering the top pipelines usually bear comparable level of risks. A case study on a real gas pipeline network consisting of more than 6500 pipelines verifies the effectiveness of the proposed model.

1. Introduction

Urban lifelines which include power, gas, water supply sewerage and other systems play important roles in maintaining city function. As a complex network distributed in the cities, the lifeline systems consisting of a great quantity of pipelines, have the characteristics of high publicity, strong correlation and great risk. The accidents in the pipeline network may cause enormous economic loss as well as serious threats to individuals, society and environment. On the other hand, the daily operation and maintenance management of the lifeline systems is associated with many aspects of data, including reports, records, logs achieved by manual inspection, pressure and flow data in Supervisory Control And Data Acquisition (SCADA) system, vector data of pipeline geographical information system (GIS) system, as well as the environmental data alongside the pipeline. How to use these large-scale and multi-source data to evaluate the severity of consequences in advance in the case that an accident, like gas or water leakage, happens in the urban pipeline network (UPN) is of great significance for risk control and resource deployment during the management and maintenance process of lifeline systems.

At present, there are three main methods for pipeline post-event risk assessment: qualitative model, semi-quantitative model and quantitative model. Through a large number of case studies, the qualitative analysis method enables to find out the degree of influence of various factors on pipeline failure, and finally formulate pipeline prevention and control measures according to the consequences of the accidents [1,2]. This method is characterized by its simplicity and practicability. The semi-quantitative analysis method is the most widely used method over the past few decades. It was first systematically proposed in the Pipeline Risk Management Manual [3] from the view of engineering application in long-distance oil and gas pipelines, and was improved in 2004 [4]. In this method, some influencing factors are selected based on the experts' experience and assigned to different weights, and then added up to obtain a total risk score representing the relative risk of the pipeline. Afterwards there are many extensions to the semi-quantitative model: some optimize the weight distribution to reduce subjectivity and apply the model to different specific scenarios [5,6]; the others employ the advanced mathematical models like Analytic Hierarchy Process (AHP), fuzzy comprehensive evaluation, event tree and fault tree [7–10] to achieve this goal. Taking into account the uncertainty of risk factors, the

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Table 1
Check of consistency.

Primary indicators	Secondary indicators	Variable	Quantification and classification Criteria	Score (10 points)	
First primary indicator used for AHP Weight=0.5	Characteristics of the in-pipe transport medium	/	Classification	1. Natural gas 2. Coal gas 3. Liquefied petroleum gas	9 8 8
	The environmental conditions alongside the pipeline	Type of buildings	Classification	1. Critical facilities 2. Commercial buildings 3. Industrial buildings 4. Residential houses	10 8 9 7
			Quantification	The number of each type of the buildings within the statistical area* of the pipeline	Each type of building's number multiply corresponding score, then add up to get B_i
		Population density	Quantification	The number of population within the statistical area* of the pipeline	$\frac{Y_i}{Y_{\max}} \times 10$
Second primary indicator used for AHP weight=0.5	Characteristics of the pipeline	Pressure	Quantification	Directly exported from the database of the GIS and SCADA system	$\frac{P_i}{P_{\max}} \times 10$
		Diameter	Quantification		$1 - \frac{D_i}{D_{\max}} \times 10$
		Thickness	Quantification		$1 - \frac{C_i}{C_{\max}} \times 10$
		Age	Quantification		$\frac{N_i}{N_{\max}} \times 10$
		Buried depth	Quantification		$1 - \frac{H_i}{H_{\max}} \times 10$
		Pipe length	Quantification		See Table 3
		Material	Classification	1. Steel 2. Plastic	10 5
		Laying method	Classification	1.Exposed 2.Buried	See Table 3

*The statistical area is determined according to Table 2.

quantitative analysis methods are further proposed, which calculate the probability of pipeline failure and combine the quantified failure consequence severity to obtain the pipeline risk value. For instance, through directed acyclic graphs, Bayesian networks [11–13] represent a group of random variables and their conditional correlations, while the bow-tie model [14] combines Fault Tree Analysis (FTA) and Event Tree Analysis (ETA) methods to analyze the whole process of an accident as well as the links between its parts. In addition to the two methods, there are other probabilistic methods, such as those based on cumulative fatal length [15], multidimensional risk evaluation based on Monte Carlo simulation [16], failure prediction based on historical accidents [17], system dynamics [18] and risk management based on Representational State Transfer (REST) technology [19].

It should be noted that these three methods all have limitations. The qualitative models or semi-quantitative models depend more or less on the estimation and experience of experts. Although the semi-quantitative models use mathematical tools to improve the objectivity of the evaluation results, it is essentially based on experts' experience and vision, which makes it difficult to apply to different and various situations. The quantitative models can accurately calculate the probability of different types of failures and the consequences of failures, thereby supporting further risk analysis of UPN. However, most quantitative models show over-dependence on division and calculation of probability events and ignore the existing data of pipelines, which may seriously influence the performance of the models. In view of all the above situations, the data-driven models emerged to solve the challenges.

As a powerful tool to classify raw unlabeled data sets, unsupervised clustering algorithms have been used in pipeline risk assessment. Zhang [20] established a pipeline risk assessment method based on principal component analysis (PCA) and clustering and applied it to the segmented assessment of a gas pipeline. Zhang et al [21] combined two methods Mutual Information Principal Component Analysis (MIPCA) and Weighted Aggregated Sum Product Assessment (WASPAS) to justify the weights of safety factors. Lim et al. [22] adopted spectral clustering algorithm to divide a lifeline network is into an adequate number of clusters, and the reduced size of the simplified network enables the

selective recursive decomposition algorithm (S-RDA) to perform the network risk assessment efficiently. Zhu and Zou [23] used hierarchical clustering to optimize the selection of artificial neural network (ANN) learning samples, overcoming the one-sidedness of relying on expert experience to predict. Although these methods usually use clustering algorithms as the preprocessing step to divide samples into groups, which simplifies the manual risk rating, they all treat the pipelines in the network as isolated individuals for analysis, ignoring the geographic or spatial relations among them. However, in engineering practice, pipelines that share adjacent joints or spatial locations usually bear comparable level of risks. Therefore, it may be helpful to introduce the topological relations of UPN to further enhance the performance of the risk assessment model. In nature, UPN is a network consisting of joints/vertices and pipelines/edges, which can be simplified into a node graph for subsequent analysis. In that case, one popular clustering method named graph clustering can come in handy. Schaeffer [24] first systematically defined and elaborated graph clustering, presented global algorithms for producing a clustering for the entire vertex set of an input graph. To implement graph clustering, Tian et al. [25] proposed a simple method, which first learns a non-linear embedding of the original graph by stacked autoencoders (AE), and then runs k-means on the embedding to obtain clustering result. Ng [26] introduced the sparse autoencoder learning algorithm to automatically learn features from unlabeled data. Henceforward, many works in graph clustering are proposed, continuing to improve the algorithm and expand the application field. The technique of DeepWalk [27], LINE [28], Node2vec [29], GraphAttention [30] were successively used in the field of Network embedding or Graph embedding. Furthermore, Xie et al. [31] proposed Deep Embedded Clustering (DEC), a method that simultaneously learns feature representations and cluster assignments using deep neural networks. More recently, Graph Convolutional Network (GCN) [30–32] was adopted for this field to do the task of node classification, graph classification, link prediction and so on. The above methods aimed at extracting topology features from networks, which can be employed to extract geographical features to improve the performance of a clustering model.

This work proposes a data-driven model for the post-event risk assessment on the accident consequence of UPN. The main novelty is

Table 2
Statistical area.

Pressure(MPa) × diameter(mm)	Length (m)	Width (m)
Less than 10	100	15
Between 10 and 20	100	30
Greater than 20	100	50

that the model uses the attribute features and topological features of a pipeline network to analyze while taking advantage of some advanced tools including clustering and GCN. The rest of this paper is arranged as follows: In Section 2, the factors affecting accident consequence of gas UPN are first summarized. The traditional assessment model based on expert scoring method and Analytical Hierarchy Process (AHP) is briefly introduced in Section 3. A new data-driven model is then proposed in Section 4 based on graph embedding and deep clustering, which employs both attribute and geographical features of the UPN. A case study on a real urban gas pipeline network is demonstrated in Section 5, where the two methods are performed and compared given the multi-source data sets and analysis results. Finally, Section 6 summarizes this work's highlights and discusses further direction.

2. Factors affecting accident consequence of gas UPN

Based on engineering practice and literature research [4,33,34], the factors which affect the accident consequence of gas leakage in UPN can be divided into three categories: the characteristics of the in-pipe transport medium, the characteristics of the pipeline itself, and the surrounding environment along the pipeline. All the items are listed in Table 1 and described in details as follows.

2.1. Characteristics of the in-pipe transport medium

For the pipelines responsible for the transport of various mediums, such as gas, oil, drinking or waste water and so on, the consequences and affects caused by the leakage can be significantly different since each fluid has the unique flammable, explosive, poisonous or deleterious nature. As this study focuses on gas pipeline network, the in-pipe mediums only include three types: natural gas, coal gas, and liquefied petroleum gas considering the real-time and long-term hazards they might impose on.

2.2. Characteristics of the pipeline

The influencing factors regarding the characteristics of the pipeline include eight indicators: six quantification items for pressure, diameter, thickness, age, length and buried depth; two classification items for pipe material and laying method. Usually, these data can be directly acquired from the database of the GIS and SCADA system.

2.3. Environmental conditions alongside the pipeline

The surrounding environments alongside the pipeline including the conditions of buildings and populations have significant impacts on the consequence which an accident of gas pipeline might cause. For buildings, they are divided into four types: critical facilities, commercial, industrial and residential buildings. The number of each type of

buildings and the populations within a certain zone nearby the pipeline are counted. The statistical area is set for every hundred meters of pipeline with the distance away from the pipeline given in Table 2 according to different diameters and pressures of the pipelines [34].

The subscript i in the variable represents the score of the i th pipe segment, the subscript max represents the highest score of all pipe segments in the UPN for this factor.

3. Traditional assessment model

3.1. Overview

Firstly, the attribute data of the pipelines needs to be obtained and evaluated if adequate. Then the raw data will be preprocessed to make them uniform and precise. According to the Kent index method [3,4], the final risk value is the sum of the scores of each indicator as shown in Table 1, ignoring the difference in the importance of each indicator. Therefore, the analytic hierarchy process (AHP) is introduced to determine the weight of each indicator, and finally multiply with corresponding score and add up to get a total risk value, thus obtain the risk ranking. The logic flow of the model is shown in Fig. 1 below. The Kent index method and AHP will be described below.

3.2. Data preprocessing

To improve the traditional assessment model's performance, data preprocessing is applied in the multisource data from UPN which aims to make the pipelines' features uniform and precise.

3.2.1. Quantification of the categorical attributes

Preprocessing is required for categorical attributes, because the clustering algorithm can only accept the numerical data. For the Boolean variables, we can replace True with 1, and False with 0. Similarly, for attributes with two or three categories, we can also assign 1, 2 or 1, 2, 3, respectively, such as the attributes of transportation medium and laying method mentioned above.

In most cases, domain knowledge is very useful, which can help us carry out some feature conversion when facing with attributes with more than three categories. For example, pipes are made of different materials, such as steel, plastic, and concrete. Compared to simply encoding it as an ordered integer (such as 1, 2, 3...), it is more reasonable to encode them as their service life because the service life reflects the durability of the material. However, this method may not be completely appropriate since the material has other properties apart from durability, which may affect the final result.

3.2.2. Re-encoding of the numerical attributes

Some attributes are numeric but not properly formatted, such as diameter, length, age, etc. Taking the diameter as an example, it is not clear whether the larger one is safer or the opposite. Nevertheless, domain knowledge is also very useful here, Wang and Li [35] used historical fault data provided by the *European Gas Pipeline Incident Data Group* (EGIG) to determine the relationship between diameter and failure frequency. The weighted diameter was proved to have a higher correlation with the occurrence of faults. Besides, the diameter conversion of the square tube is considered. Assuming that the section size of a square tube is $a \times b$, the equivalent diameter $\frac{2ab}{(a+b)}$ is used to keep

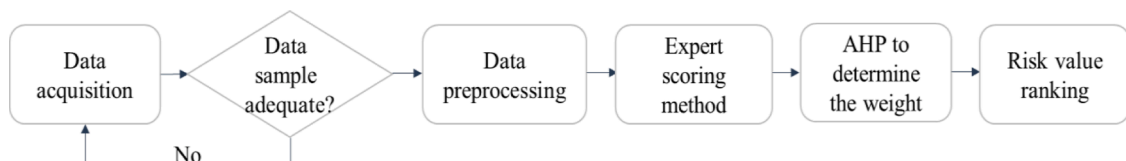


Fig. 1. Flowchart of the traditional risk-assessment method.

Table 3
Score of the combined index of pipe length and laying method.

Length Laying method	0–10 m	10–25 m	25–50 m	> 50m
Exposed	1	2	3	5
Buried	6	7	8	10

consistent with the diameter of the round tube.

3.3. Expert scoring based on the Kent index method

Since the data of the influencing factors have been obtained, the data can be processed to obtain the relative risk. This article mainly refers to some methods in the existing literatures [4,33–35], organizes and summarizes a set of scoring system as shown in Table 1. In the risk assessment, the evaluation value of each secondary indicator score is between 0 and 10 points, the evaluation value is multiplied by the index weight, and all indicators are added together to obtain the risk assessment value of the gas pipeline. Here are some points that needs to be explained.

(1) This study focuses on urban gas pipeline networks, the in-pipe transport medium is all municipal gas. Therefore, there is no difference in scoring.

(2) In the characteristic indexes of the pipeline itself, the pipeline pressure, diameter, age (service life), wall thickness, minimum buried depth, pipe length, laying method and material can be obtained based on the basic data of gas pipeline network laying and operation [36]. Considering the positive and negative correlation between the data value and the risk, the score method for a gas pipeline network containing n pipelines is set in Table 1.

(3) The pipe length L and laying method are combined as an index to give a score, see Table 3. The material failure can be evaluated according to the laying method and the length of pipeline. The longer the gas pipeline is, the more it will be affected by the stress caused by the long-term chronic movement of the soil, and the greater the possibility of pipeline failure and crack. And if the pipeline is buried, it is easier to be affected by the movement of soil and thus cause failure compared to exposed.

3.4. Analytical hierarchy process to determine the weight of the indexes

Analytic hierarchy process (AHP) [37,38] is a systematic and quantitative method for the decision-making process of complex phenomena. It is mainly used to determine the relative importance of many influencing factors. The main steps include establishing hierarchical model, constructing pairwise comparison matrix, calculating weight vector and checking consistency.

The hierarchical model of AHP consists of two primary indicators, which contain 3 and 6 secondary indicators, respectively. As shown in Table 1, the first primary indicator contains the characteristics of the in-pipe transport medium and the environmental conditions alongside the

pipeline, including transport medium, type (B) and number (Y) of buildings and population density. The other primary indicator contains the characteristics of the pipeline, including pipeline pressure (P), pipe diameter (D), pipe age (N), material (M), buried depth (H), pipeline length and laying method combined as (L).

To determine the weights of the indicators in the AHP, a paired comparison matrix should be introduced for the indicators at the same level. For the two primary indicators, they are considered as equally important and the comparison matrix M_1 is given in Eq. (1). For the first set of secondary indicators, since the transport medium for all pipe sections is generally the same and not considered as a variable, the type (B) and number (Y) of buildings and population density are assumed as of equal importance, which gives the comparison matrix M_{21} shown in Eq. (1). As for the second set of secondary indicators, a 6×6 paired comparison matrix is introduced as shown in Eq. (2) by comparing the significance of each two indicators based on the industry experience.

$$M_1 = M_{21} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad (1)$$

$P \quad D \quad N \quad M \quad H \quad L$

$$M_{22} = \begin{bmatrix} 1 & 2 & 4 & 5 & 5 & 3 \\ 1/2 & 1 & 2 & 3 & 3 & 2 \\ 1/4 & 1/2 & 1 & 2 & 2 & 1/2 \\ 1/5 & 1/3 & 1/2 & 1 & 1 & 1/2 \\ 1/5 & 1/3 & 1/2 & 1 & 1 & 1/3 \\ 1/3 & 1/2 & 2 & 2 & 3 & 1 \end{bmatrix} \begin{matrix} P \\ D \\ N \\ M \\ H \\ L \end{matrix} \quad (2)$$

For a pairwise comparison matrix, the maximum eigenvalue and corresponding eigenvector are calculated, and the consistency is tested by Random Consistency Ratio (CR). If the test passes, the feature vector (after normalization) is the weight vector; If not, the pairwise comparison matrix needs to be reconstructed. Generally, for the judgment matrix of order $n \geq 3$, when $CR \leq 0.1$, the consistency of the judgment matrix is considered acceptable. For M_{22} , $CR=0.0130 < 0.1$, so it passed the consistency test.

Following the process proposed by Saaty [38], the weights of all the influencing factors can be then calculated. For a certain pipe section, the score of each influencing factor is multiplied by the corresponding weight and then added up to get a total score, which can be used as the index for further risk ranking.

4. Data-driven assessment model

4.1. Overview

As mentioned in Section 1, although the traditional risk assessment model adopts the analytic hierarchy process to reduce the subjectivity of expert ratings, its subjectivity still exists. Given this, a fully data-driven model is proposed for post-event risk assessment on UPN. The flowchart of the method is presented in Fig. 2, which consists of four steps: data

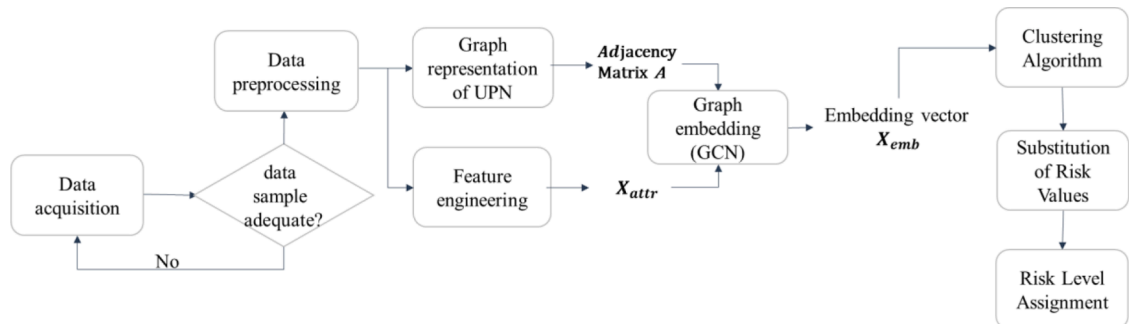


Fig. 2. Flowchart of the proposed data-driven assessment method.

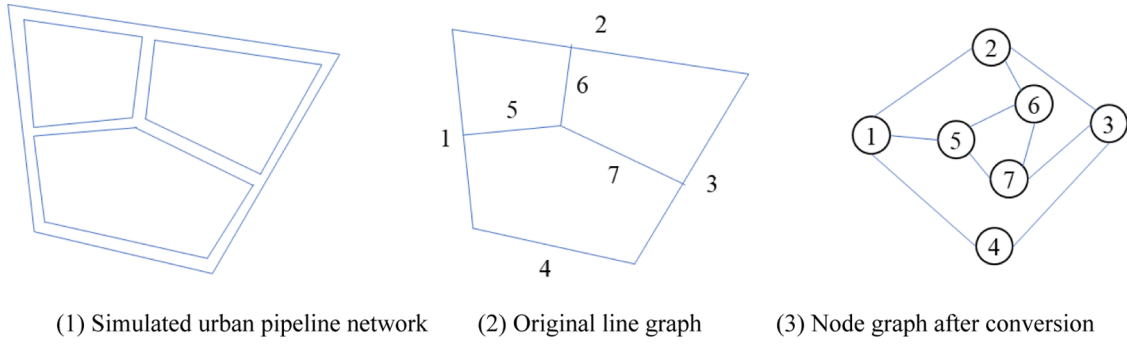


Fig. 3. network graph conversion.

acquisition and preprocessing, feature extraction, clustering and risk level assignments. Feature extraction is a key step to extract the most representative features from the raw data sets, promoting the model's capability in clustering. In this work, two types of features are employed: one is X_{attr} which is the pipelines' attributes after feature engineering; the other is adjacency matrix A , which is the topological features extracted from the graph representation of UPN. Using the embedding feature X_{emb} obtained from GCN, the pipelines with similar features are grouped into one cluster by clustering algorithms and the risk levels are assigned according to the average risk value of each cluster obtained in Section 3.

4.2. Feature engineering for X_{attr} extraction

Using the preprocessed data acquired in Section 3.2, a highly condensed feature vector X_{attr} representing the pipelines' attributes can be extracted via feature scaling.

Due to the difference in the magnitude of various features, the raw data need to be normalized before feeding to the clustering model. A linear normalization method called Min-Max scaling is adopted here:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (3)$$

where x is the raw data; x' is the normalized data; $\max(x), \min(x)$ are the maximum and minimum values of the original data. This method can map the raw data to the range [0–1].

4.3. Graph embedding for X_{emb} extraction

Graph Embedding is also called network representation learning. The main purpose is to map the nodes in a network to a low-dimensional vector based on the characteristics of the network, so that the similarity among the nodes can be quantitatively measured. In graph embedding, an undirected graph $G(V, E)$ is constructed based on the original topology of UPN, where the vertex V represents a node or a valve, and the edge E represents a pipeline. In the actual graph embedding algorithm, feature extraction is mainly focused on the nodes, so the original network needs to be converted. The basic procedure for network conversion is illustrated in Fig. 3. To convert a line graph into a node graph, the pipelines are first numbered. If starting from one line, there is a way to get to another line through certain intersection, then their corresponding nodes are also connected [39]. Through this method, node-centric model can be obtained, while retaining the relationship between the pipelines.

Graph Convolutional Neural Network (GCN), has become a research hotspot in the field of deep learning in recent years. Compared with Convolutional Neural Network (CNN) which can only handle more regular European data, GCN is more powerful in dealing with irregular and complicated graph-structured data and can efficiently implement node classification.

Fig. 4 presents the overall framework of a GCN. Given a graph with node features, via several hidden layers, a lower dimensional feature matrix Z can be achieved which contains the topological feature of the graph.

Suppose the graph of a UPN consists of N nodes and each node has its own D types of characteristics. The characteristics of these nodes are

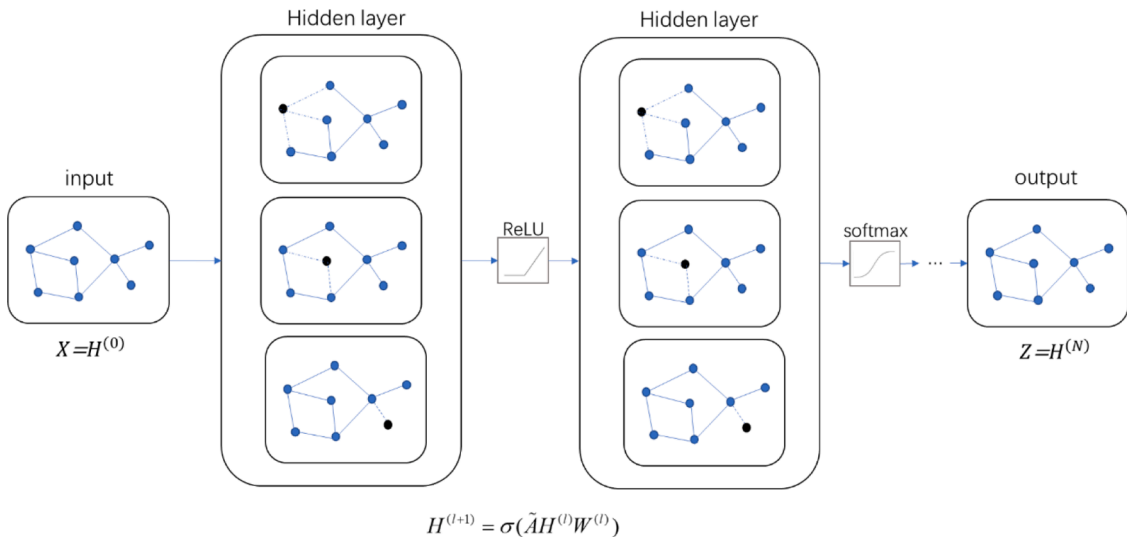


Fig. 4. Framework of GCN.

formed into the $N \times D$ input feature matrix X ; the relations between every two nodes (adjacent-1; not adjacent-0) are formed into the $N \times N$ adjacency matrix A . Taking X and A as inputs, the propagation formula of GCN between layers is:

$$H^{(l+1)} = \sigma(\tilde{A}H^{(l)}W^{(l)}) \quad (4)$$

Where:

$$\tilde{A} = \hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}}$$

$\hat{A} = A + I$, I is the Identity matrix; $\hat{D}$ is the degree matrix of $\hat{A}$, $\hat{D}_{ii} = \sum_j \hat{A}_{ij}$;

H is the features of each layer, for the input layer, H is X ;

σ is a nonlinear activation function;

W is the weight matrix that needs training.

If a two-layer GCN is constructed as shown in Fig. 4, LeakyReLU is used as the activation functions to reduce information loss, and the formula for forward propagation is:

$$Z = \text{Softmax}(\tilde{A}_{\text{LeakyReLU}}(\tilde{A}_{XW^{(0)}})W^{(1)}) \quad (5)$$

By training the weight matrix W via inverse propagation, the optimal Z can be achieved, which is actually the low-dimensional graph embedding representation X_{emb} . GCN considers the features of the pipelines connected to the pipeline itself and incorporate them with its own features. In this case, it's an unsupervised learning process to capture the embedding vector for each pipeline to distinguish them with the input features.

4.4. Clustering algorithms

In combination with the pipelines' attribute X_{attr} via feature engineering and the topological feature A via UPN, a fused feature vector X_{emb} can be constructed and used as the inputs for further clustering algorithms. The core task lies in the selection and implementation of the appropriate clustering algorithms considering the size and characteristics of the datasets in a specific application scenario. In this work, three methods are employed and compared: clustering based on segmentation, clustering based on hierarchy, and spectral clustering, which will be highlighted next.

4.4.1. Clustering based on segmentation

Segmentation-based clustering treats the samples as scattered points. Given the number of target clusters, the initial center points are first selected, and iteratively reset according to the pre-selected heuristic algorithm until the intra-class points are close enough and the inter-class points are far enough. Among them, the K-means method is the most common clustering algorithm. The task of the algorithm is to cluster the datasets x into k clusters based on the minimization loss function:

$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|_2^2 \quad (6)$$

$$\mu_i = \frac{1}{|C_i|} \sum_{x \in C_i} x \quad (7)$$

where μ_i is the center point of the cluster C_i . The smaller the value of WCSS (within-cluster sum of squares), the more similar the samples in the cluster.

4.4.2. Hierarchical clustering

Hierarchical clustering can be further divided into a bottom-up merge type and a top-down split type. The former one is used here. It starts from the bottom layer, and each time the closest points are grouped to form the cluster of the upper layer. When a certain termi-

nation condition is reached or when all data points are merged into a cluster, it stops. The distance of the clusters commonly used in hierarchical clustering is measured as the average distance:

$$d_{\text{avg}}(C_i, C_j) = \frac{1}{|C_i||C_j|} \sum_{x \in C_i} \sum_{z \in C_j} \text{dist}(x, z) \quad (8)$$

where the distance function (x, z) can be selected as required, and the Euclidean distance is commonly used.

4.4.3. Spectral clustering

Spectral clustering is a clustering method based on graph theory. It clusters the eigenvectors of Laplacian matrix of sample data, so as to achieve the purpose of clustering sample data. Spectral clustering can be understood as mapping data from high-dimensional space to low-dimensional space, and then clustering in low-dimensional space by other clustering algorithms (such as K-means) [40].

4.4.4. Selection of the clustering methods

Generally speaking, the K-means algorithm is simple and efficient with low complexity in time and space, suitable for processing large dataset. However, it is very sensitive to the selection of the preset k value and the initial center point, and also sensitive to noise and outliers. In contrast, the hierarchical clustering algorithm has better interpretability, which is convenient for further labeling and classification. It can also deal with the non-spherical cluster problem that the K-means algorithm cannot solve. As a greedy algorithm based on local optimization, however, its time complexity is high ($O(n^3)$) and there may be a cascading effect caused by the deviation of the previous step leading to problems in the subsequent steps. The hierarchical clustering is suitable for clustering scenarios with small data volume and complex cluster shape, while the spectral clustering is more effective for dealing with sparse data or high dimensional data.

In addition to qualitative judgment, the Silhouette Coefficient (s) is used to quantitatively evaluate the clustering results. It is defined for each sample as:

$$s = \frac{b - a}{\max(a, b)} \quad (9)$$

in which, a is the mean distance between a sample and all the other points in the same class; b is the mean distance between a sample and all the other points in the next nearest cluster. The Silhouette Coefficient is ranged from -1 and 1. A higher Silhouette Coefficient score relates to a model with better defined clusters.

4.5. Risk level assignment

Through clustering analysis, the pipelines of similar attributes are gathered in the same cluster, so the next step is to assign risk level to the different clusters. This study takes advantage of the risk values for each pipeline obtained from the previous traditional model to solve this problem. According to the results achieved by the traditional model, the average value of the pipeline risk values in the same cluster can be calculated and compared with each other, based on which the same risk level can be assigned to the pipelines in each cluster. This method combines the results of the traditional model and the clustering algorithms.

5. Case study

5.1. Background

To demonstrate the application of the three proposed methods, a case study is conducted on a real municipal gas pipeline network. An urban area with $4 \text{ km} \times 2 \text{ km}$ around is selected as the object region, covering 6589 gas pipelines, unevenly-distributed buildings and



Fig. 5. Street architecture in the scope of the pipe network in 3D renderings.

Table 4
Risk value output.

Risk ranking	Risk value	Pipeline identifier
1	17.001	1280
2	16.365	2746
...	...	...
6599	4.942	1514

population. The original data is provided by a local Survey and Design Institute, consisting of information about the pipelines' attributes and environmental conditions alongside the pipelines exported from geographic information system (GIS) system. The data sample acquired fully represents the situation of various pipelines with different surroundings, and the data size is large enough to draw some general conclusions. The scope of the pipe network within the yellow border line and street architecture renderings of the data set are shown in Fig. 5

5.2. Results of the traditional assessment model

According to the flowchart shown in Fig. 1, the original data including nine attributes (Variables $B, Y, P, D, C, N, H, L, M$ as listed in Table 1) for 6589 pipelines is first preprocessed and converted to scores. Using the paired comparison matrixes for the two-level indicators (M_1, M_{21}, M_{22}) as described in Section 3.4, the weight vectors (W_1, W_{21}, W_{22}) can be determined as:

$$W_1 = W_{21} = [0.5 \quad 0.5] \quad (10)$$

$$W_{22} = [0.390 \quad 0.218 \quad 0.107 \quad 0.068 \quad 0.064 \quad 0.153] \quad (11)$$

After determining the weights of the indexes, the final risk assessment results can be achieved by multiplying the scoring matrix with the weight vector. The typical outputs are listed in Table 4, including risk ranking level, relative risk value, and the corresponding identifier for the all 6589 pipelines.

5.3. Results of the data-driven model

As shown in Fig. 2, the first three steps until data preprocessing are basically the same as the traditional model. Through feature scaling and compression as mentioned in Section 4.2, the feature matrix X_{attr} can be obtained. The next step is to extract X_{emb} . The spatial connection tool is implemented to get the edge list of the pipeline network diagram in ArcGIS, and reconstruct the topology graph. With the 6589×9 feature

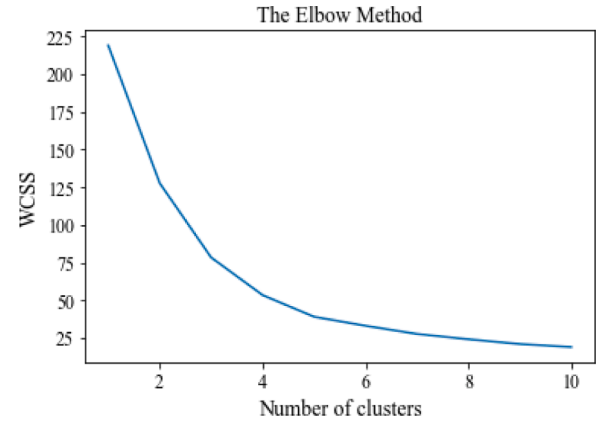


Fig. 6. Optimal number of clusters.

matrix X and the 6589×6589 adjacency matrix A , the first-layer target dimension of GCN is set as 4 and the second-layer as 2. At present, Silhouette Coefficient is used to train the weights. Since the feature matrix X is also input into the GCN layers, the result after training is already the fused feature X_{emb} that can be used for subsequent clustering.

As the clustering number k needs to be preset, the Elbow Method [41] is used to determine the most suitable k . Fig. 6 presents the relation of the preset number of clusters (k) and WCSS (see Eq. (4)) of each result. It can be seen that WCSS decreases with the value k at the outset. When $k > 5$, the increase of the number of clusters has little effect on improving the degree of distortion. The slope no longer decreases but tends to be stable after 5. Therefore $k = 5$ is taken as the last elbow.

The results of the K-means, hierarchical clustering and spectral clustering algorithms are plotted in Fig. 7. Different clusters are marked with scattered dots of different colors and different markers. It can be seen that the results of the three algorithms are quite different. Table 5 gives the quantitative test of the clustering results by Silhouette Coefficient mentioned in Section 4, where it can be clearly seen the performance of K-means is the best which has above 0.5 Silhouette Coefficient. Thus K-means is best suited for this application scenario. Considering the effectiveness and computational efficiency, the K-means method is recommended.

According to the method proposed in Section 4.5, the risk level is evaluated. The 6589 pipe samples are divided into five clusters with their respective number of sample points, average risk value and risk

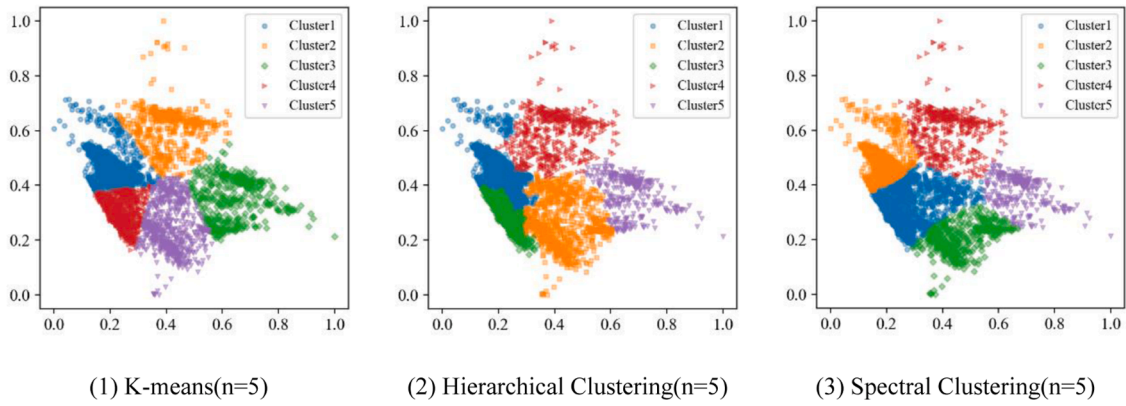


Fig. 7. Clustering results of different algorithms.

Table 5

Quantitative test of clustering results.

Clustering algorithms	K-means	Hierarchical Clustering	Spectral Clustering
Silhouette Coefficient (s)	0.501	0.466	0.438
Performance	best	good	worst

Table 6

Risk level assignment.

Cluster number	1	2	3	4	5
Number of sample points	773	2372	510	2440	494
Average risk value	8.260	7.298	8.940	7.751	9.143
Median risk value	7.827	6.945	8.741	7.469	9.204
Risk rank	3	1	4	2	5
Risk level	Mid	Low	Mid-high	Mid-low	High

level. The results are shown in Table 6 below. A bigger number in risk rank represents higher risk (more serious accident consequence). The differences of average risk values among clusters are significant and easy to distinguish. Lower risk clusters have much more pipe samples compared to higher risk ones according to the results.

To clearly display the difference of the distribution of risk values among the clusters, a histogram of risk value distribution in each cluster from X_{emb} is provided as shown in Fig. 8. It can be seen that the dense distribution range of each cluster is different.

5.4. Comparison and discussion

To visualize the results of the two models, the heat maps of the risk assessment results of the traditional model is presented in the pipe network, as shown in Fig. 9(1). The dark-colored line represents the pipe segment with high risk level. Here, the relative risk is divided into five levels according to the natural discontinuity method.

In contrast, the distribution of the risk levels achieved by the data-driven model in the pipe network shown in Fig. 9(2) has obvious spatial correlation, and pipelines with similar geographical locations have the similar risk levels, which is consistent with the expectations for the model. Meanwhile, long distance transport pipelines tend to have higher after-accident risk than the short and dense pipelines, which is similar to the results of the traditional model. Therefore, the data-driven model combines the geographic features and the pipeline attributes perfectly. When a whole region's pipes share the same level of risk in the map, it's more convenient to manage, operate and maintain these pipe regions rather than treating individually.

In fact, it can be intuitively seen that the results of the two models are

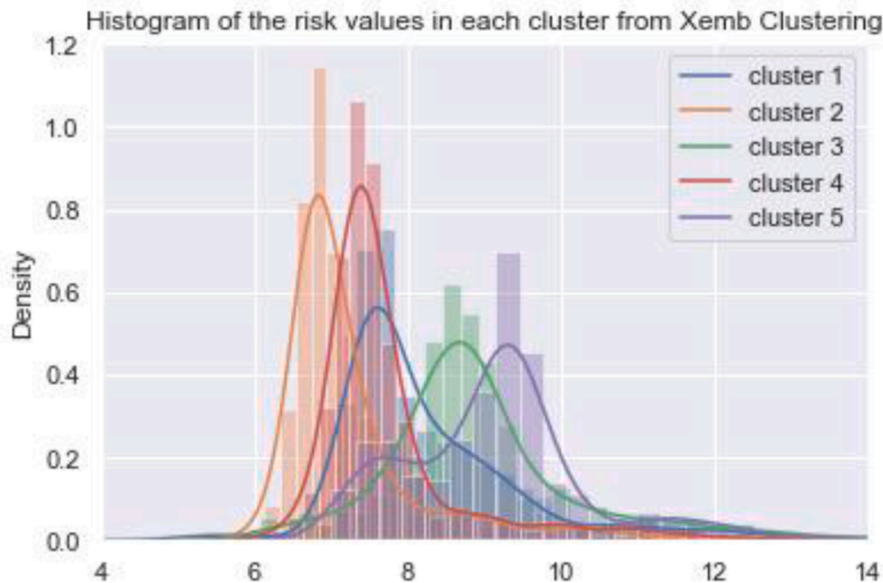
Fig. 8. Histogram of the risk values in each cluster from X_{emb} .



Fig. 9. Heat map of the risk assessment results.

quite different, because the traditional model uses natural discontinuity method to divide all pipeline segments into 5 risk levels uniformly while the clustering results of the 5 risk levels are quite uneven. Fig. 10 is the overall rendering that adds streets and buildings based on the results of the data-driven model. The pipeline regions with greater risk can be seen clearly, which has practical significance for the pre-judgment and maintenance of the actual project.

As a case study, the proposed data-driven model has been applied to an urban gas pipeline network in this research. It can be expanded to other types pipe network, like water, electricity and so on. In addition, this risk assessment can be applied to any kind of complex network system with the feature data of its basic elements. Since this model relies on computer to perform nearly all the work once the data is acquired, the computational efficiency is significantly increased compared to traditional methods.

However, the proposed model also has some limitations. First, the proposed model takes advantage of the results of risk values from traditional model to classify different clusters into different risk levels because the clustering outputs don't have labels. It's a feasible but not perfect method because the subjectivity of traditional method still exists. Second, setting a standard to divide risk level of the clustering results is

hard because the risk values can be various from case to case. But the main goal is to make the average risks among clusters have significant difference and easy to distinguish. More subsequent analyses are desired to improve this new model especially addressing these questions.

6. Conclusion

Using the multi-source data from UPN, a data-driven model is proposed based on graph embedding and deep clustering for post-event risk assessment on the accident consequence of the pipeline network. Major conclusions are summarized as follows:

- (1) This paper systematically sorts out the factors affecting accident consequence of gas UPN and establishes the post-event risk evaluation indicators. A traditional risk assessment model based on the Kent index method and the analytic hierarchy process is employed to determine the relative risk value and risk ranking of each pipeline in the network.
- (2) A new data-driven model is proposed for post-event risk assessment on UPN, which divides pipelines into a few clusters and then assigns each cluster with different risk level. This model is



Fig. 10. Overall rendering of the risk assessment results.

superior to the traditional model in less dependence on experts' subjective judgements or calculation of probability events in Bayes decision procedure. Besides, the latest GCN technology is used to extract the topological features of pipe network as a complement to the common attribute features for clustering analysis considering the adjacent pipelines usually take on comparable level of risks.

- (3) A case study is conducted on a real municipal gas pipeline network distributed in an urban area with $4 \text{ km} \times 2 \text{ km}$ around covering more than 6500 gas pipelines. The traditional model and the proposed data-driven model are employed to evaluate the post-event risk of the pipeline network. It demonstrates that the spatially connected pipelines have similar risk levels and the pipelines in densely populated areas present larger risk levels, which is consistent with expectations and verifies the effectiveness of the proposed model. The comparison of the three clustering algorithms also shows that *K*-means is the most suitable method considering the effectiveness and efficiency for this specific application scenarios.
- (4) The proposed method has good scalability and computational efficiency while there are some limitations like the output of the model can't be used to evaluate individual pipelines and hard to set a standard for classifying different clusters. So for application, the traditional method and the proposed method should be combined to use in practice. For instance, to have an overall understanding of the risk level distribution in a certain region's pipeline network, the data-driven method is suitable and convenient. And when zoom in to inspect specific pipe sections' risk or accident consequence, the traditional method will be helpful. Additionally, only relying on computer methods is not a rational choice especially in the risk assessment field. Experts' judgements are still crucial to decide risk situations while taking advantage of the data analysis results.

CRedit authorship contribution statement

Yang Yang: Methodology, Investigation, Writing – original draft.
Suzhen Li: Conceptualization, Methodology, Writing – original draft.
Pengcheng Zhang: Data curation, Investigation, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Han ZY, Weng WG. Comparison study on qualitative and quantitative risk assessment methods for urban natural gas pipeline network. *J Hazard Mater* 2011; 189:509–18. 1–2.
- [2] Cekirge HM. Qualitative risk of gas pipelines. *Am J Energy Eng* 2015;3(3):53–6.
- [3] Muhlbauer WK. Pipeline risk management manual. Houston, TX: Gulf Publishing Company; 1996.
- [4] Muhlbauer WK. Pipeline risk management manual: ideas, techniques, and resources. Elsevier; 2004.
- [5] Wang KL, Cao MM, Wang BD. Application of muhlbauer risk assessment method in pipeline risk assessment. *Res Environ Sci* 2006;19(2):112–4.
- [6] Zhang YQ, Zhao W. Application and analysis of muhlbauer assessment method in oil pipeline risk assessment. *Res Environ Sci* 2010;36(1):69–72.
- [7] Jiang DF, Zhang YT, Yue BB. Risk evaluation of town gas pipelines based on fuzzy analytic hierarchy process. *Contemp Chem Ind* 2011;40(8):873–6.
- [8] Jamshidi A, Yazdani-Chamzini A, Yakhchali SH. Developing a new fuzzy inference system for pipeline risk assessment. *J Loss Prev Process Ind* 2013;26(1):197–208.
- [9] Huang XM, Li BZ, Peng SN. Quantitative risk analysis of natural gas pipeline based on event tree. *Gas Heat* 2009;29(4):42–6.
- [10] Liang W, Hu J, Zhang L. Assessing and classifying risk of pipeline third-party interference based on fault tree and SOM. *Eng Appl Artif Intell* 2012;25(3): 594–608.
- [11] Wu WS, Yang CF, Chang JC. Risk assessment by integrating interpretive structural modeling and Bayesian network, case of offshore pipeline project. *Reliab Eng Syst Saf* 2015;142:515–24.
- [12] Zhang Y, Weng WG. Bayesian network model for buried gas pipeline failure analysis caused by corrosion and external interference. *Reliab Eng Syst Saf* 2020; 203:107089.
- [13] Zarei E, Azadeh A, Khakzad N. Dynamic safety assessment of natural gas stations using Bayesian network. *J Hazard Mater* 2017;321:830–40.

- [14] Hao MJ, You QJ, Yue Z. Risk analysis of urban gas pipeline network based on improved bow-tie model. *IOP Conf Ser Earth Environ Sci* 2017;93(1):012059.
- [15] Jo YD, Ahn BJ. A method of quantitative risk assessment for transmission pipeline carrying natural gas. *J Hazard Mater* 2005;123(1–3):1–12.
- [16] Medeiros CP, Alencar MH, de Almeida AT. Multidimensional risk evaluation of natural gas pipelines based on a multicriteria decision model using visualization tools and statistical tests for global sensitivity analysis. *Reliab Eng Syst Saf* 2017;165:268–76.
- [17] Azari P, Karimi M. Quantitative risk mapping of urban gas pipeline networks using GIS. *International archives of the photogrammetry. Remote Sens Spat Inf Sci* 2017; XLII-4/W4:319–24.
- [18] Liu A, Chen K, Huang X. Dynamic risk assessment model of buried gas pipelines based on system dynamics. *Reliab Eng Syst Saf* 2021;208:107326.
- [19] Tang R, Bai Y, Wang T. Research and application of risk assessment in urban lifeline systems based on GIS and REST. In: *Proceedings of the second world congress on software engineering*. 1. IEEE; 2010. p. 248–51.
- [20] Zhang J. Pipeline risk assessment method based on principle component-clustering analysis. *Oil Gas Storage Transp* 2014;33(2):139–43.
- [21] Zhang H, Dong S, Ling J. A modified method for the safety factor parameter: the use of big data to improve petroleum pipeline reliability assessment. *Reliab Eng Syst Saf* 2020;198:106892.
- [22] Lim HW, Song J, Kurtz N. Seismic reliability assessment of lifeline networks using clustering-based multi-scale approach. *Earthq Eng Struct Dyn* 2015;44(3):355–69.
- [23] Zhu L, Zou L. Application of cluster analysis neural network in the safety evaluation for the urban underground gas pipeline. In: *Proceedings of the 6th world congress on intelligent control and automation*. 1. IEEE; 2006. p. 2845–9.
- [24] Schaeffer SE. Graph clustering. *Comput Sci Rev* 2007;1(1):27–64.
- [25] Tian F, Gao B, Cui Q. Learning deep representations for graph clustering. In: *Proceedings of the Twenty-eighth AAAI conference on artificial intelligence*; 2014.
- [26] Ng ANG, editor. Sparse autoencoder. CS294A Lecture notes, 72. Stanford University; 2011. p. 1–19.
- [27] Perozzi B, Al-Rfou R, Skiena S. Online learning of social representations. In: *Proceedings of the 20th ACM SIGKDD international conference on knowledge discovery and data mining*; 2014. p. 701–10.
- [28] Tang J, Qu M, Wang M. Line: large-scale information network embedding. In: *Proceedings of the 24th international conference on world wide web*; 2015. p. 1067–77.
- [29] Grover A, Leskovec J. node2vec: scalable feature learning for networks. In: *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*; 2016. p. 855–64.
- [30] Veličković P, Cucurull G, Casanova A. Graph attention networks. *ArXiv* 2018;abs/1710.10903.
- [31] Xie J, Girshick R, Farhadi A. Unsupervised deep embedding for clustering analysis. In: *Proceedings of the international conference on machine learning*; 2016. p. 478–87.
- [32] Hu F, Zhu Y, Wu S, et al. Semi-supervised Node Classification via Hierarchical Graph Convolutional Networks. *ArXiv* 2019;abs/1902.06667.
- [33] National Energy Administration. Oil and gas pipeline risk assessment method part 1: semi-quantitative assessment method. Beijing: China Standard Press; 2012. sy/t 6891.1-2012.
- [34] Yang YC. Evaluation of consequences severity in safety assessment for town gas piping. *Chem Equip Pipeline* 2005;42(6):22–5.
- [35] Wang Z, Li S. Data-driven risk assessment on urban pipeline network based on a cluster model. *Reliab Eng Syst Saf* 2020;196:106781.
- [36] Han ZY. Research on the risk assessment method of urban gas pipeline network. Beijing: Tsinghua University; 2010. in Chinese.
- [37] Golden BL, Wasil EA, Harker PT. The analytic hierarchy process. applications and studies. Berlin, Heidelberg: Springer; 1989.
- [38] Saaty TL. Decision making with the analytic hierarchy process. *Int J Serv Sci* 2008;1(1):83–98.
- [39] Crucitti P, Latora V, Porta S. The network analysis of urban streets: a dual approach. *Phys A Stat Mech Appl* 2006;369(2):853–66.
- [40] Von Luxburg U. A tutorial on spectral clustering. *Stat Comput* 2007;17(4):395–416.
- [41] Kodinariya TM, Makwana PR. Review on determining number of cluster in K-means clustering. *Int J* 2013;1(6):90–5.